Chen 2020 — Nasopharyngeal Carcinoma (NPC)

EBV-associated NPC · 10x · 3,674 malignant cells (8 samples) · embryonic-program cell-state analysis

Verdict

Pattern does NOT reproduce — Gate 1 NO-GO. The malignant compartment clusters by patient, not into a sample-integrated continuous manifold, so no trajectory is fit.

Parameters (derived, not inherited)

min-cell threshold = 62 (median-restricted spillover gap 107→36, ×2.97; naive top-gap kept only 2 samples — rejected). Overdispersed genes = 1,290 (GAM). n_pcs = 6 (scree elbow). Harmony θ=2 (balance of the θ=1–3 sweep).

Gate 1 — NO-GO (mixed? NO / continuous? yes — both required)

Continuous (1 connected component, 100% of cells) BUT not mixed: across the integration-strength sweep 73%→56%→1% of cells sit in patient-pure Leiden clusters (θ=1/2/3); diff-sample neighbour fraction 0.30 at θ=2 vs 0.69 ceiling. The purest clusters (99% P11) are NON-cycling — genuine patient-private CNV states, not a shared axis. P11+P01 alone = 72% of cells. θ=3 forces mixing only by over-integrating (a detached speck appears). Do not force a trajectory.

Gates 2 / 3

Not evaluated — gate chain stops at Gate 1 NO-GO (no manifold / signatures / correlation / modules notebooks produced).

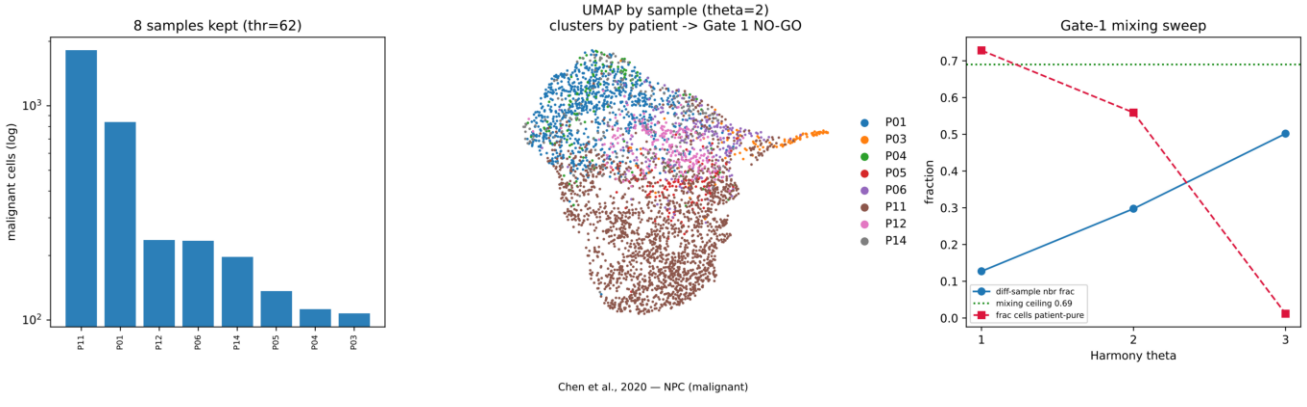
Hub / r

N/A — no manifold, no hub.

EMT-TFs (unconditional step, descriptive only)

Detection-gated panel (≥3%): PRRX1 (62%) › SNAI2 (24%) › FOXC1 (13%) › ZEB2 (13%) › ZEB1 (7%) › TWIST2 (6%). Canonical SNAI1 (1.6%) & TWIST1 (1%) fall below the floor — a Snail/Zeb/Twist-only panel would miss the mesenchymal PRRX/FOXC signal.

01 · Preprocessing — per-sample counts, patient-segregated UMAP, mixing sweep (samples.pdf)



02 · EMT-TF expression overlay — descriptive, patient-structured embedding (expression.pdf)

